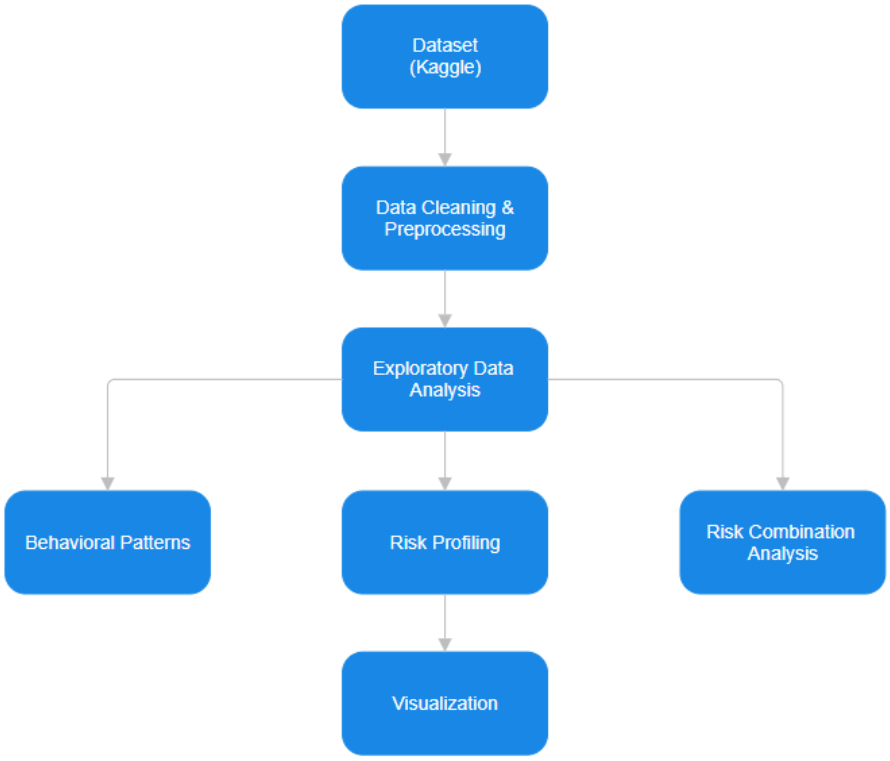
1. Problem Statement

Financial fraud in digital transactions has become a major challenge for banks and online payment platforms. Fraudulent activities can cause financial losses and undermine trust in digital financial systems.

The goal of this project is to analyze transaction data and identify behavioral patterns associated with fraudulent transactions. By examining factors such as device usage, payment methods, transaction types, and user behavior, the project aims to uncover insights that can help detect suspicious activities and support the development of fraud monitoring systems.

2. Solution Architecture

The solution follows a data analytics pipeline designed to identify fraud patterns through exploratory analysis and behavioral risk profiling.



3. Tech Stack

Programming

- Python

Data Analysis

- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn

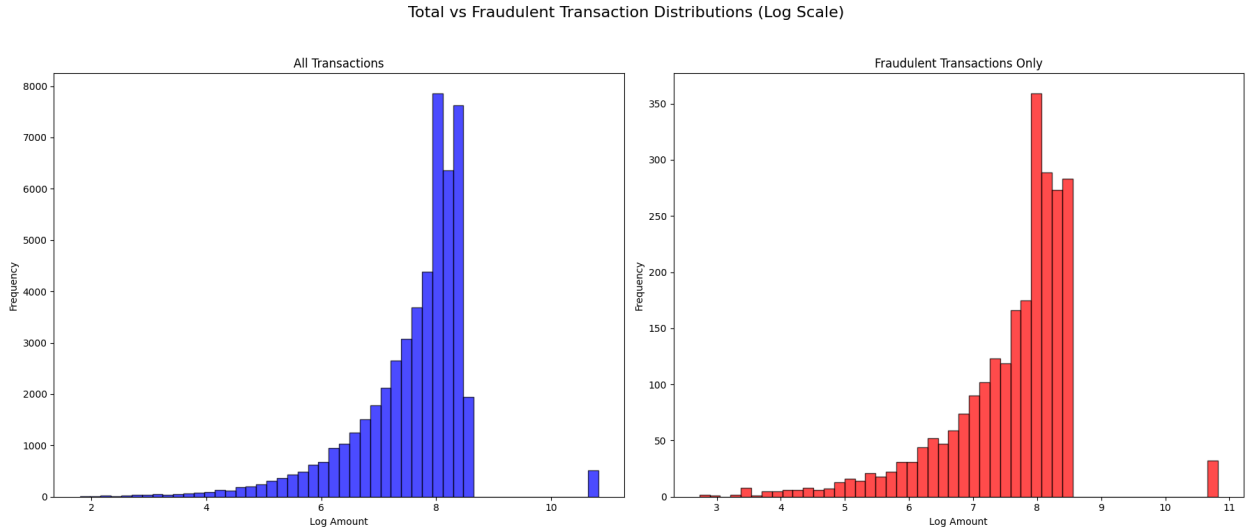
Dashboard / Presentation

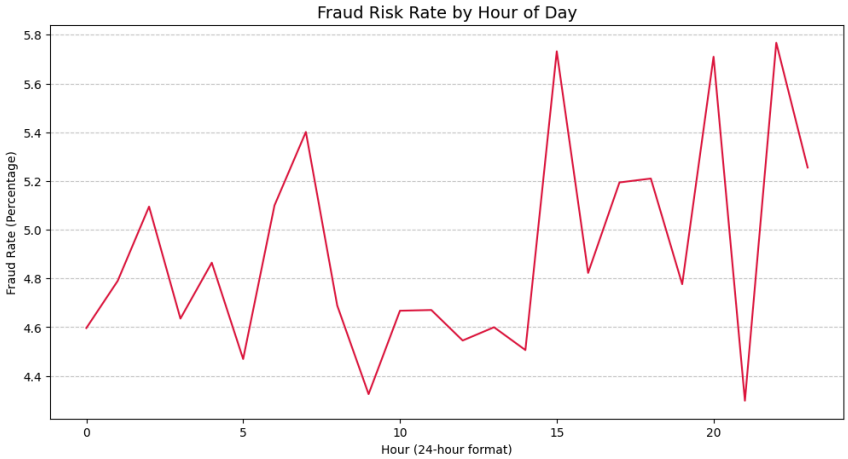
- PowerPoint Dashboard

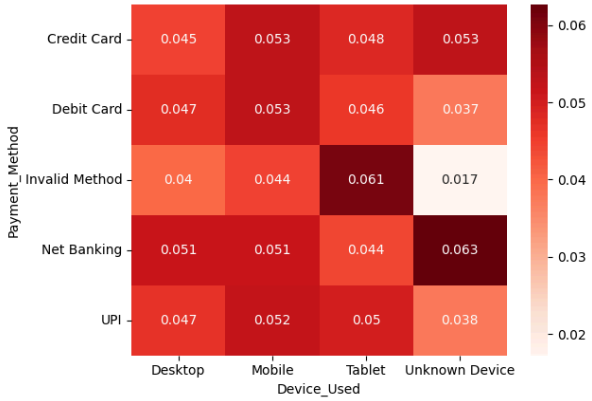
Dataset Source

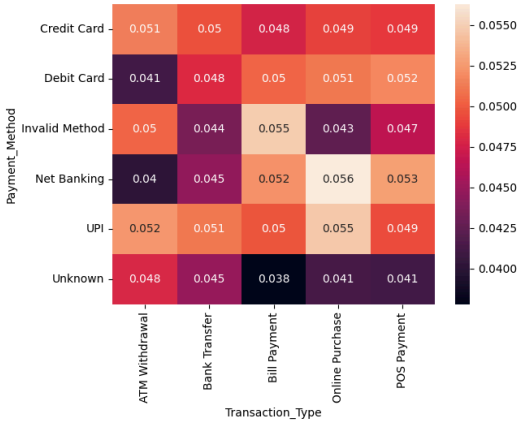
- Kaggle Dataset ([Fraud Detection Dataset]()

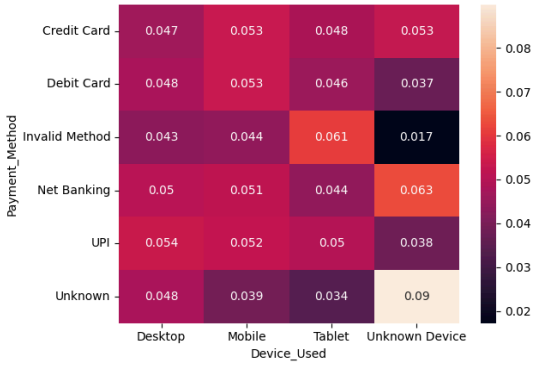
4. Screenshots




Fraud Risk Rate by Hour of Day
Fraud Rate (Percentage)
Hour (24-hour format)


Payment_Method
Device_Used
Credit Card 0.045 0.053 0.048 0.053
Debit Card 0.047 0.053 0.046 0.037
Invalid Method 0.04 0.044 0.061 0.017
Net Banking 0.051 0.051 0.044 0.063
UPI 0.047 0.052 0.05 0.038
Desktop
Mobile
Tablet
Unknown Device


Payment_Method
Transaction_Type
Credit Card 0.051 0.05 0.048 0.049 0.049
Debit Card 0.041 0.048 0.05 0.051 0.052
Invalid Method 0.05 0.044 0.055 0.043 0.047
Net Banking 0.04 0.045 0.052 0.056 0.053
UPI 0.052 0.051 0.05 0.055 0.049
Unknown 0.048 0.045 0.038 0.041 0.041
ATM Withdrawal
Bank Transfer
Bill Payment
Online Purchase
POS Payment


Payment_Method
Device_Used
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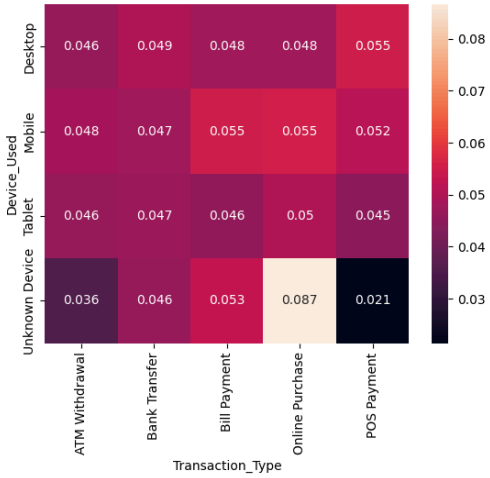
Debit Card 0.048 0.053 0.046 0.037
Invalid Method 0.043 0.044 0.061 0.017
Net Banking 0.05 0.051 0.044 0.063
UPI 0.054 0.052 0.05 0.038
Unknown 0.048 0.039 0.034 0.09
Desktop
Mobile
Tablet
Unknown Device


Device_Used
Transaction_Type
Desktop 0.046 0.049 0.048 0.048 0.055
Mobile 0.048 0.047 0.055 0.055 0.052
Tablet 0.046 0.047 0.046 0.05 0.045
Unknown Device 0.036 0.046 0.053 0.087 0.021
ATM Withdrawal
Bank Transfer
Bill Payment
Online Purchase
POS Payment

5. Demo Instructions

Follow these steps to run the project.

Step 1: Open the Notebook

Open the Google Colab notebook containing the analysis.

Step 2: Install Required Libraries

pip install pandas seaborn matplotlib kagglehub streamlit

Step 3: Load Dataset

The dataset is automatically downloaded using KaggleHub.

Step 4: Run the Analysis

Execute the notebook cells to generate visualizations and fraud

6. Future Improvements

Several improvements can enhance the system further:

1. Machine Learning Fraud Detection

Implement classification models such as Random Forest, XGBoost, or Neural Networks to predict fraud probability.

2. Real-Time Fraud Monitoring

Develop a real-time monitoring system capable of detecting suspicious transactions as they occur.

3. Advanced Behavioral Analysis

Incorporate additional features such as geolocation anomalies and device fingerprinting.

4. Risk Scoring System

Build a fraud risk scoring model that assigns a probability score to each transaction.

5. Interactive Analytics Platform

Expand the dashboard to include more advanced filters, real-time insights, and interactive exploration tools.